



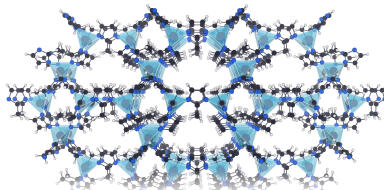
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Optical Response Properties - Tutorial

Coupled Perturbed Hartree-Fock and Kohn-Sham method

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MSSC2025
TORINO 8-12 SEP 2025

A problem to be solved

The perturbative part of the Hamiltonian for a crystal in a constant electric field contains the

$$\text{perturbative operator} \quad \hat{\Omega}(\mathbf{r}, \mathbf{k}) = \mathbf{r} + i \nabla_{\mathbf{k}} = i e^{i \mathbf{k} \cdot \mathbf{r}} \nabla_{\mathbf{k}} e^{-i \mathbf{k} \cdot \mathbf{r}}$$

Problem : How to derive the form of the perturbative operator?

Find the classical Hamiltonian for a particle in external electromagnetic field

Starting point. Form of the Lorentz force : $\mathbf{F} = \frac{d}{dt}(m\mathbf{v}) = q \left[\mathcal{E}(\mathbf{r}, t) + \frac{1}{c} \mathbf{v} \wedge \mathcal{B}(\mathbf{r}, t) \right]$

Step one. The form of the Lagrangian $\mathcal{L}_{\text{em}}$ can be deduced comparing the form of the Lorentz force with the Euler Lagrange equations

$$\text{Euler Lagrange equations :} \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}_{\text{em}}}{\partial \dot{r}_{\eta}} \right) = \frac{\partial \mathcal{L}_{\text{em}}}{\partial r_{\eta}} \quad \text{where} \quad \mathcal{L}_{\text{em}} \equiv \mathcal{L}_{\text{em}}(\mathbf{r}, \dot{\mathbf{r}}, t)$$

Step two. The form of the Hamiltonian $\mathcal{H}_{\text{em}}$ can be found applying a Legendre transform to the Lagrangian

$$\text{Legendre transform} \quad \mathcal{L}_{\text{em}}(\mathbf{r}, \dot{\mathbf{r}}, t) \rightarrow \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t) \quad : \quad \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t) = p_{\eta} \dot{r}_{\eta} - \mathcal{L}_{\text{em}}(\mathbf{r}, \dot{\mathbf{r}}, t)$$

Maxwell equations

Maxwell equations expressed using Gaussian units

electric field $\mathcal{E}(\mathbf{r}, t)$

magnetic field $\mathcal{B}(\mathbf{r}, t)$

$$\nabla \cdot \mathcal{E}(\mathbf{r}, t) = 4\pi\rho$$

$$\nabla \cdot \mathcal{B}(\mathbf{r}, t) = 0$$

$$\nabla \wedge \mathcal{E}(\mathbf{r}, t) = -\frac{1}{c} \frac{\partial \mathcal{B}(\mathbf{r}, t)}{\partial t}$$

$$\nabla \wedge \mathcal{B}(\mathbf{r}, t) = \frac{1}{c} \left[4\pi\mathbf{J} + \frac{\partial \mathcal{E}(\mathbf{r}, t)}{\partial t} \right]$$

Given a vector field $\mathcal{A}(\mathbf{r}, t)$ and a scalar field $\phi(\mathbf{r}, t)$, the following vector identities hold

$$\nabla \cdot \nabla \wedge \mathcal{A}(\mathbf{r}, t) = 0$$

$$\nabla \wedge \nabla \phi(\mathbf{r}, t) = 0$$

Using the first vector identity in conjunction with the first Maxwell equation for the magnetic field leads to the possibility to express the magnetic field as

$$\nabla \cdot \mathcal{B}(\mathbf{r}, t) = 0 \quad \rightarrow \quad \mathcal{B}(\mathbf{r}, t) = \nabla \wedge \mathcal{A}(\mathbf{r}, t)$$

Using the previously derived equation for the magnetic field in the second Maxwell equation for the electric field, and then using the second vector identity, leads to the possibility to express the electric field as

$$\nabla \wedge \left[\mathcal{E}(\mathbf{r}, t) + \frac{1}{c} \frac{\partial \mathcal{A}(\mathbf{r}, t)}{\partial t} \right] = 0 \quad \rightarrow \quad \mathcal{E}(\mathbf{r}, t) = -\nabla \phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \mathcal{A}(\mathbf{r}, t)}{\partial t}$$

Classical Hamiltonian for a particle in external electromagnetic field

The previously derived equations for the electric and magnetic field as functions of the vector potential $\mathcal{A}(\mathbf{r}, t)$ and the scalar potential $\phi(\mathbf{r}, t)$, respectively, are given by

$$\mathcal{E}(\mathbf{r}, t) = -\nabla\phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \mathcal{A}(\mathbf{r}, t)}{\partial t} \quad \text{and} \quad \mathcal{B}(\mathbf{r}, t) = \nabla \wedge \mathcal{A}(\mathbf{r}, t)$$

Using these expressions for the electric and magnetic fields in the Lorentz force law leads to

$$\mathbf{F} = \frac{d}{dt}(m\mathbf{v}) = q \left[\mathcal{E}(\mathbf{r}, t) + \frac{1}{c} \mathbf{v} \wedge \mathcal{B}(\mathbf{r}, t) \right] = q \left[-\nabla\phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \mathcal{A}(\mathbf{r}, t)}{\partial t} + \frac{1}{c} \mathbf{v} \wedge \nabla \wedge \mathcal{A}(\mathbf{r}, t) \right]$$

The goal is to express the Lorentz force in a way such that the terms involving the Lagrangian $\mathcal{L}_{\text{em}}$ appearing in the Euler Lagrange equations can be recognized, that is, to rewrite all the terms with expressions containing only the time derivative d/dt and spatial derivatives $\nabla_{\mathbf{r}} \equiv \nabla$

By applying a general vector identity to the last term in the previous equation, it becomes

$$\mathbf{a} \wedge (\mathbf{b} \wedge \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} \quad \rightarrow \quad \mathbf{v} \wedge \nabla \wedge \mathcal{A}(\mathbf{r}, t) = \nabla [\mathbf{v} \cdot \mathcal{A}(\mathbf{r}, t)] - (\mathbf{v} \cdot \nabla) \mathcal{A}(\mathbf{r}, t)$$

so that the Lorentz force assumes the form

$$\mathbf{F} = \frac{d}{dt}(m\mathbf{v}) = q \left\{ -\nabla\phi(\mathbf{r}, t) - \frac{1}{c} \left[\frac{\partial \mathcal{A}(\mathbf{r}, t)}{\partial t} + (\mathbf{v} \cdot \nabla) \mathcal{A}(\mathbf{r}, t) \right] + \frac{1}{c} \nabla [\mathbf{v} \cdot \mathcal{A}(\mathbf{r}, t)] \right\}$$

Classical Hamiltonian for a particle in external electromagnetic field

$$\mathbf{F} = \frac{d}{dt}(m\mathbf{v}) = q \left\{ -\nabla\phi(\mathbf{r}, t) - \frac{1}{c} \left[\frac{\partial \mathcal{A}(\mathbf{r}, t)}{\partial t} + (\mathbf{v} \cdot \nabla) \mathcal{A}(\mathbf{r}, t) \right] + \frac{1}{c} \nabla [\mathbf{v} \cdot \mathcal{A}(\mathbf{r}, t)] \right\}$$

The term on the left hand side contains a total time derivative d/dt as searched. Moreover, the first and last terms in the right hand side of the previous equation can be collected as functions of the spatial derivatives ∇ only. The remaining terms to fix are the ones collected in the square parenthesis in the right hand side of the equation. These can be rewritten as a total time derivative in the following way

$$\frac{d}{dt} \mathcal{A}(\mathbf{r}, t) = \frac{\partial \mathcal{A}(\mathbf{r}, t)}{\partial t} + \frac{dr_\eta}{dt} \frac{\partial}{\partial r_\eta} \mathcal{A}(\mathbf{r}, t) = \frac{\partial \mathcal{A}(\mathbf{r}, t)}{\partial t} + (\mathbf{v} \cdot \nabla) \mathcal{A}(\mathbf{r}, t)$$

where the Einstein notation is used. Inserting this expression in the Lorentz force law leads to

$$\mathbf{F} = \frac{d}{dt}(m\mathbf{v}) = q \left\{ -\nabla\phi(\mathbf{r}, t) - \frac{1}{c} \frac{d}{dt} \mathcal{A}(\mathbf{r}, t) + \frac{1}{c} \nabla [\mathbf{v} \cdot \mathcal{A}(\mathbf{r}, t)] \right\}$$

Finally, collecting all the terms that depend on the total time derivative d/dt and on the spatial derivatives ∇ gives the form

$$\frac{d}{dt} \left[m\mathbf{v} + \frac{q}{c} \mathcal{A}(\mathbf{r}, t) \right] = \nabla \left[-q\phi(\mathbf{r}, t) + \frac{q}{c} \mathbf{v} \cdot \mathcal{A}(\mathbf{r}, t) \right]$$

In components :

$$\frac{d}{dt} \left[m v_\eta + \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \right] = \frac{\partial}{\partial r_\eta} \left[-q\phi(\mathbf{r}, t) + \frac{q}{c} v_\xi \mathcal{A}_\xi(\mathbf{r}, t) \right]$$

Classical Hamiltonian for a particle in external electromagnetic field

The previous equation can be rewritten using the relation $v_\eta = \dot{r}_\eta$ as

$$\frac{d}{dt} \left[m \dot{r}_\eta + \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \right] = \frac{\partial}{\partial r_\eta} \left[-q \phi(\mathbf{r}, t) + \frac{q}{c} \dot{r}_\xi \mathcal{A}_\xi(\mathbf{r}, t) \right]$$

and comparing this form with the Euler Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}_{\text{em}}}{\partial \dot{r}_\eta} \right) = \frac{\partial \mathcal{L}_{\text{em}}}{\partial r_\eta} \quad \text{where} \quad \mathcal{L}_{\text{em}} \equiv \mathcal{L}_{\text{em}}(\mathbf{r}, \dot{\mathbf{r}}, t)$$

the form of the Lagrangian can be found to be

$$\mathcal{L}_{\text{em}} = \frac{1}{2} m \dot{r}_\eta \dot{r}_\eta + \frac{q}{c} \dot{r}_\xi \mathcal{A}_\xi(\mathbf{r}, t) - q \phi(\mathbf{r}, t) = \frac{1}{2} m v_\eta v_\eta + \frac{q}{c} v_\xi \mathcal{A}_\xi(\mathbf{r}, t) - q \phi(\mathbf{r}, t)$$

The conjugate momentum with respect to the variable r_η is defined as

$$p_\eta = \frac{\partial \mathcal{L}_{\text{em}}}{\partial \dot{r}_\eta} = m \dot{r}_\eta + \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) = m v_\eta + \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \quad \rightarrow \quad v_\eta = \frac{1}{m} \left[p_\eta - \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \right]$$

Finally, using the Legendre transform $\mathcal{L}_{\text{em}}(\mathbf{r}, \dot{\mathbf{r}}, t) \rightarrow \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t)$ the Hamiltonian is

$$\mathcal{H}_{\text{em}} = p_\eta \dot{r}_\eta - \mathcal{L}_{\text{em}} = \left[m v_\eta + \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \right] v_\eta - \frac{1}{2} m v_\eta v_\eta - \frac{q}{c} v_\xi \mathcal{A}_\xi(\mathbf{r}, t) + q \phi(\mathbf{r}, t)$$

$$\mathcal{H}_{\text{em}} \equiv \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t) = \frac{1}{2m} \left[\mathbf{p} - \frac{q}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + q \phi(\mathbf{r}, t)$$

Classical Hamiltonian for a particle in external electromagnetic field

The previous equation can be rewritten using the relation $v_\eta = \dot{r}_\eta$ as

$$\frac{d}{dt} \left[m \dot{r}_\eta + \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \right] = \frac{\partial}{\partial r_\eta} \left[-q \phi(\mathbf{r}, t) + \frac{q}{c} \dot{r}_\xi \mathcal{A}_\xi(\mathbf{r}, t) \right]$$

and comparing this form with the Euler Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}_{\text{em}}}{\partial \dot{r}_\eta} \right) = \frac{\partial \mathcal{L}_{\text{em}}}{\partial r_\eta} \quad \text{where} \quad \mathcal{L}_{\text{em}} \equiv \mathcal{L}_{\text{em}}(\mathbf{r}, \dot{\mathbf{r}}, t)$$

the form of the Lagrangian can be found to be

$$\mathcal{L}_{\text{em}} = \frac{1}{2} m \dot{r}_\eta \dot{r}_\eta + \frac{q}{c} \dot{r}_\xi \mathcal{A}_\xi(\mathbf{r}, t) - q \phi(\mathbf{r}, t) = \frac{1}{2} m v_\eta v_\eta + \frac{q}{c} v_\xi \mathcal{A}_\xi(\mathbf{r}, t) - q \phi(\mathbf{r}, t)$$

The conjugate momentum with respect to the variable r_η is defined as

$$p_\eta = \frac{\partial \mathcal{L}_{\text{em}}}{\partial \dot{r}_\eta} = m \dot{r}_\eta + \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) = m v_\eta + \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \quad \rightarrow \quad v_\eta = \frac{1}{m} \left[p_\eta - \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \right]$$

Finally, using the Legendre transform $\mathcal{L}_{\text{em}}(\mathbf{r}, \dot{\mathbf{r}}, t) \rightarrow \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t)$ the Hamiltonian is

$$\mathcal{H}_{\text{em}} = \frac{1}{2} m v_\eta v_\eta + q \phi(\mathbf{r}, t) = \frac{1}{2m} \left[p_\eta - \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \right] \left[p_\eta - \frac{q}{c} \mathcal{A}_\eta(\mathbf{r}, t) \right] + q \phi(\mathbf{r}, t)$$

$$\mathcal{H}_{\text{em}} \equiv \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t) = \frac{1}{2m} \left[\mathbf{p} - \frac{q}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + q \phi(\mathbf{r}, t)$$

Classical Hamiltonian for a particle in external electromagnetic field

$$\mathcal{H}_{\text{em}} \equiv \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t) = \frac{1}{2m} \left[\mathbf{p} - \frac{q}{c} \mathcal{A}(\mathbf{r}, t) \right]^2 + q \phi(\mathbf{r}, t)$$

To simplify the expression of the Hamiltonian, a convenient gauge can be chosen, that defines the form for the vector potential $\mathcal{A}(\mathbf{r}, t)$, namely

$$\text{Length gauge :} \quad \mathcal{A}(\mathbf{r}, t) = 0 \quad \rightarrow \quad \mathcal{H}_{\text{em}} \equiv \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t) = \frac{\mathbf{p}^2}{2m} + q \phi(\mathbf{r}, t)$$

Note that, in the length gauge, the magnetic field is zero, so that the rotor of the electric field is also zero, due to the second Maxwell equation for the electric field, that is

$$\text{Length gauge :} \quad \mathcal{A}(\mathbf{r}, t) = 0 \quad \rightarrow \quad \mathcal{B}(\mathbf{r}, t) = 0 \quad \rightarrow \quad \nabla \wedge \mathcal{E}(\mathbf{r}, t) = 0$$

Therefore, supposing an expression for the electric field given by a plane wave with the form

$$\mathcal{E}(\mathbf{r}, t) = \mathcal{E}_0 e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$$

it can be shown that the length gauge restricts the solution of the Maxwell equations to longitudinal electric field, with the direction of propagation $\hat{\mathbf{q}} \equiv \mathbf{q}/\|\mathbf{q}\|$ of the electromagnetic wave parallel to the electric field pointing direction, that is

$$\nabla \wedge \mathcal{E}(\mathbf{r}, t) = \nabla e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} \wedge \mathcal{E}_0 = \mathbf{q} \wedge \mathcal{E}_0 e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} = 0 \quad \Longleftrightarrow \quad \mathbf{q} \wedge \mathcal{E}_0 = 0$$

$$\Longleftrightarrow \quad \mathbf{q} \parallel \mathcal{E}_0 \quad \rightarrow \quad \mathcal{E}(\mathbf{r}, t) = \mathcal{E}_0 \hat{\mathbf{q}} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} \quad \text{so that} \quad \mathcal{E}_0 = \mathcal{E}_0 \hat{\mathbf{q}}$$

Classical Hamiltonian for a particle in external electromagnetic field

Length gauge : $\mathcal{A}(\mathbf{r}, t) = 0 \quad \rightarrow \quad \mathcal{H}_{\text{em}} \equiv \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t) = \frac{\mathbf{p}^2}{2m} + q \phi(\mathbf{r}, t)$

To find the form of the scalar potential $\phi(\mathbf{r}, t)$, the relation previously derived expressing the electric field in terms of the scalar and vector potential can be used in the length gauge, namely

$$\mathcal{E}(\mathbf{r}, t) = -\nabla\phi(\mathbf{r}, t) - \frac{1}{c} \frac{\partial \mathcal{A}(\mathbf{r}, t)}{\partial t} \quad \text{in the length gauge :} \quad \mathcal{E}(\mathbf{r}, t) = -\nabla\phi(\mathbf{r}, t)$$

so that the expression for the scalar potential $\phi(\mathbf{r}, t)$ can be derived from the last relation, knowing the form of the electric field, in the following way

$$\phi(\mathbf{r}, t) = - \int \mathcal{E}(\mathbf{r}, t) \cdot d\mathbf{r} \quad \text{with} \quad \mathcal{E}(\mathbf{r}, t) = \mathcal{E}_0 \hat{\mathbf{q}} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$$

Carrying out the line integral of the electric field leads the form

$$\phi(\mathbf{r}, t) = -\mathcal{E}_0 \int \hat{\mathbf{q}} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} \cdot d\mathbf{r} = \frac{i\mathcal{E}_0}{q} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} + \text{constant} \quad \text{where} \quad q = \|\mathbf{q}\|$$

and taking the constant equal to $-i\mathcal{E}_0/q$ gives the form

$$\phi(\mathbf{r}, t) = \frac{i\mathcal{E}_0}{q} \left[e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} - 1 \right]$$

Classical Hamiltonian for a particle in external electromagnetic field

So far, the motion of a charged particle in an external electromagnetic field has been described. The Hamiltonian can be easily modified if, in addition to the external fields just treated, the particle experiences a conservative potential, as the Coulomb potential $U(\mathbf{r})$ that is unrelated to the external electromagnetic field. In this case, the total Hamiltonian form is

$$\mathcal{H}(\mathbf{r}, \mathbf{p}, t) = \mathcal{H}_{\text{em}}(\mathbf{r}, \mathbf{p}, t) + U(\mathbf{r}) = \frac{\mathbf{p}^2}{2m} + U(\mathbf{r}) + q \phi(\mathbf{r}, t) = \mathcal{H}_0(\mathbf{r}, \mathbf{p}) + \mathcal{H}_\phi(\mathbf{r}, t)$$

$$\text{where } \mathcal{H}_0(\mathbf{r}, \mathbf{p}) = \frac{\mathbf{p}^2}{2m} + U(\mathbf{r}) \quad \text{and} \quad \mathcal{H}_\phi(\mathbf{r}, t) = q \phi(\mathbf{r}, t) = \frac{iq\mathcal{E}_0}{q} \left[e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} - 1 \right]$$

where q and m are respectively the charge and the mass of the particle, $\mathbf{p}$ is the electron momentum, $U(\mathbf{r})$ the Coulomb potential energy without outer fields, $\phi(\mathbf{r}, t)$ is a scalar potential derived using the length gauge and $\mathcal{E}_0$ is the intensity of the electric field

For an electron ($q = -e$) in a static field ($\omega = 0$)

$$\mathcal{H}(\mathbf{r}, \mathbf{p}) = \mathcal{H}_0(\mathbf{r}, \mathbf{p}) + \mathcal{H}_\phi(\mathbf{r})$$

$$\text{where } \mathcal{H}_0(\mathbf{r}, \mathbf{p}) = \frac{\mathbf{p}^2}{2m} + U(\mathbf{r}) \quad \text{and} \quad \mathcal{H}_\phi(\mathbf{r}) = -\frac{ie\mathcal{E}_0}{q} (e^{i\mathbf{q} \cdot \mathbf{r}} - 1)$$

Hamiltonian matrix elements

For a static field ($\omega = 0$) : $\mathcal{H}_\phi(\mathbf{r}) = -\frac{ie\mathcal{E}_0}{q} (e^{i\mathbf{q}\cdot\mathbf{r}} - 1) \rightarrow \hat{\mathcal{H}}_\phi = -\frac{ie\mathcal{E}_0}{q} (e^{i\mathbf{q}\cdot\mathbf{r}} - 1)$

To evaluate the matrix elements for the Hamiltonian term $\hat{\mathcal{H}}_\phi$ this operator can be applied on the wavefunction $\psi_n(\mathbf{r}, \mathbf{k})$ written according to the Bloch theorem as the product of a plane wave $e^{i\mathbf{k}\cdot\mathbf{r}}$ times a function $u_n(\mathbf{r}, \mathbf{k})$ that is periodic with the periodicity of the lattice

$$\psi_n(\mathbf{r}, \mathbf{k}) = u_n(\mathbf{r}, \mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}$$

The matrix elements of the Hamiltonian operator term $\hat{\mathcal{H}}_\phi$ can be computed as

$$\begin{aligned} \hat{\mathcal{H}}_\phi^{nm}(\mathbf{k}, \bar{\mathbf{k}}) &= -\frac{ie\mathcal{E}_0}{q} \left[\int_{-\infty}^{\infty} d\mathbf{r} \psi_n^*(\mathbf{r}, \mathbf{k}) e^{i\mathbf{q}\cdot\mathbf{r}} \psi_m(\mathbf{r}, \bar{\mathbf{k}}) - \int_{-\infty}^{\infty} d\mathbf{r} \psi_n^*(\mathbf{r}, \mathbf{k}) \psi_m(\mathbf{r}, \bar{\mathbf{k}}) \right] \\ \int_{-\infty}^{\infty} d\mathbf{r} \psi_n^*(\mathbf{r}, \mathbf{k}) e^{i\mathbf{q}\cdot\mathbf{r}} \psi_m(\mathbf{r}, \bar{\mathbf{k}}) &= \int_{-\infty}^{\infty} d\mathbf{r} u_n^*(\mathbf{r}, \mathbf{k}) e^{i(\bar{\mathbf{k}}-\mathbf{k}+\mathbf{q})\cdot\mathbf{r}} u_m(\mathbf{r}, \bar{\mathbf{k}}) \\ &= \sum_{\ell \in \mathbb{Z}} e^{i(\bar{\mathbf{k}}-\mathbf{k}+\mathbf{q})\cdot\ell\mathbf{a}} \int_0^{\mathbf{a}} d\mathbf{r} u_n^*(\mathbf{r}, \mathbf{k}) e^{i(\bar{\mathbf{k}}-\mathbf{k}+\mathbf{q})\cdot\mathbf{r}} u_m(\mathbf{r}, \bar{\mathbf{k}}) \\ \text{using the relation } \sum_{\ell \in \mathbb{Z}} e^{i(\bar{\mathbf{k}}-\mathbf{k}+\mathbf{q})\cdot\ell\mathbf{a}} &= \frac{2\pi}{\mathbf{a}} \delta(\bar{\mathbf{k}} - \mathbf{k} + \mathbf{q}) \\ &= \frac{2\pi}{\mathbf{a}} \delta(\bar{\mathbf{k}} - \mathbf{k} + \mathbf{q}) \int_0^{\mathbf{a}} d\mathbf{r} u_n^*(\mathbf{r}, \mathbf{k}) e^{i(\bar{\mathbf{k}}-\mathbf{k}+\mathbf{q})\cdot\mathbf{r}} u_m(\mathbf{r}, \bar{\mathbf{k}}) \end{aligned}$$

The above integral consists of zero terms except those obeying the equality $\bar{\mathbf{k}} = \mathbf{k} - \mathbf{q}$

Hamiltonian matrix elements

In order to consider only the non zero matrix elements, the condition $\bar{\mathbf{k}} = \mathbf{k} - \mathbf{q}$ can be substituted in the first form of the perturbative Hamiltonian matrix elements, thus leading to

$$\begin{aligned}\hat{\mathcal{H}}_{\phi}^{nm}(\mathbf{k}, \mathbf{k} - \mathbf{q}) &= -\frac{i e \mathcal{E}_0}{q} \left[\int_{-\infty}^{\infty} d\mathbf{r} \psi_n^*(\mathbf{r}, \mathbf{k}) e^{i\mathbf{q} \cdot \mathbf{r}} \psi_m(\mathbf{r}, \mathbf{k} - \mathbf{q}) - \int_{-\infty}^{\infty} d\mathbf{r} \psi_n^*(\mathbf{r}, \mathbf{k}) \psi_m(\mathbf{r}, \mathbf{k} - \mathbf{q}) \right] \\ &= -\frac{i e \mathcal{E}_0}{q} \left[\int_{-\infty}^{\infty} d\mathbf{r} \psi_n^*(\mathbf{r}, \mathbf{k}) e^{i\mathbf{q} \cdot \mathbf{r}} \psi_m(\mathbf{r}, \mathbf{k} - \mathbf{q}) - \delta(\mathbf{q}) \delta_{nm} \right]\end{aligned}$$

Optical wavelengths ($100 \text{ nm} \leq \lambda \leq 800 \text{ nm}$) are at least 10^3 times larger than the average distance between atomic oscillators in matter (about 0.1 nm), which means that $\mathbf{q} \rightarrow \mathbf{0}$ if compared to $\mathbf{k}$ so that it is allowed to take the limit for small wave vector, that is

$$\hat{\mathcal{H}}_{\phi}^{nm}(\mathbf{k}) = \lim_{\mathbf{q} \rightarrow \mathbf{0}} \hat{\mathcal{H}}_{\phi}^{nm}(\mathbf{k}, \mathbf{k} - \mathbf{q}) = -\lim_{\mathbf{q} \rightarrow \mathbf{0}} \left\{ \frac{i e \mathcal{E}_0}{q} \left[\int_{-\infty}^{\infty} d\mathbf{r} \psi_n^*(\mathbf{r}, \mathbf{k}) e^{i\mathbf{q} \cdot \mathbf{r}} \psi_m(\mathbf{r}, \mathbf{k} - \mathbf{q}) - \delta(\mathbf{q}) \delta_{nm} \right] \right\}$$

In this case the wavefunction $\psi_n(\mathbf{r}, \mathbf{k} - \mathbf{q})$ can be expanded as a Taylor series truncated to the first order around $\mathbf{k}$ as

$$\psi_n(\mathbf{r}, \mathbf{k} - \mathbf{q}) \simeq \psi_n(\mathbf{r}, \mathbf{k}) - \mathbf{q} \cdot \nabla_{\mathbf{k}} \psi_n(\mathbf{r}, \mathbf{k})$$

and similarly for the exponential operator $e^{i\mathbf{q} \cdot \mathbf{r}} \simeq 1 + i\mathbf{q} \cdot \mathbf{r}$

Substituting these approximations in the perturbative Hamiltonian matrix element form, it becomes

$$\hat{\mathcal{H}}_{\phi}^{nm}(\mathbf{k}) \simeq e \mathcal{E}_0 \int_{-\infty}^{\infty} d\mathbf{r} \psi_n^*(\mathbf{r}, \mathbf{k}) \hat{\Omega}(\mathbf{r}, \mathbf{k}) \psi_m(\mathbf{r}, \mathbf{k}) \quad \text{where} \quad \boxed{\hat{\Omega}(\mathbf{r}, \mathbf{k}) = \mathbf{r} + i \nabla_{\mathbf{k}}}$$

Vector Perturbation Operator

The perturbative part of the Hamiltonian operator for a crystal in a constant electric field is

$$\hat{\Omega}(\mathbf{r}, \mathbf{k}) = \mathbf{r} + i\nabla_{\mathbf{k}}$$

Another expression for this operator can be found by applying it to a wavefunction $\psi_n(\mathbf{r}, \mathbf{k})$ written according to the Bloch theorem as

$$\psi_n(\mathbf{r}, \mathbf{k}) = u_n(\mathbf{r}, \mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}$$

Evaluating the action of the perturbation operator on the Bloch wavefunction leads the expression

$$\hat{\Omega}(\mathbf{r}, \mathbf{k}) \psi_n(\mathbf{r}, \mathbf{k}) = \mathbf{r} \psi_n(\mathbf{r}, \mathbf{k}) + i\nabla_{\mathbf{k}} \psi_n(\mathbf{r}, \mathbf{k})$$

In particular, the last term can be rewritten as

$$\begin{aligned} \nabla_{\mathbf{k}} \psi_n(\mathbf{r}, \mathbf{k}) &= e^{i\mathbf{k}\cdot\mathbf{r}} \nabla_{\mathbf{k}} u_n(\mathbf{r}, \mathbf{k}) + i\mathbf{r} u_n(\mathbf{r}, \mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} = e^{i\mathbf{k}\cdot\mathbf{r}} \nabla_{\mathbf{k}} u_n(\mathbf{r}, \mathbf{k}) + i\mathbf{r} \psi_n(\mathbf{r}, \mathbf{k}) \\ &= e^{i\mathbf{k}\cdot\mathbf{r}} \nabla_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}} e^{i\mathbf{k}\cdot\mathbf{r}} u_n(\mathbf{r}, \mathbf{k}) + i\mathbf{r} \psi_n(\mathbf{r}, \mathbf{k}) \\ &= e^{i\mathbf{k}\cdot\mathbf{r}} \nabla_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}} \psi_n(\mathbf{r}, \mathbf{k}) + i\mathbf{r} \psi_n(\mathbf{r}, \mathbf{k}) \\ &= \left[e^{i\mathbf{k}\cdot\mathbf{r}} \nabla_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}} + i\mathbf{r} \right] \psi_n(\mathbf{r}, \mathbf{k}) \quad \rightarrow \quad \nabla_{\mathbf{k}} = i\mathbf{r} + e^{i\mathbf{k}\cdot\mathbf{r}} \nabla_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}} \end{aligned}$$

Substituting back the form of the gradient operator with respect to the reciprocal space vector $\mathbf{k}$ into the expression of the perturbation operator, it can be rewritten as

$$\boxed{\hat{\Omega}(\mathbf{r}, \mathbf{k}) = \mathbf{r} + i\nabla_{\mathbf{k}} = i e^{i\mathbf{k}\cdot\mathbf{r}} \nabla_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}}}$$

The Coupled Perturbed Theory

The total energy E of the perturbed system can be expressed as the sum of an unperturbed energy with the perturbed part, ruled by the vector perturbation operator, that is

$$\begin{aligned} E &= \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) \hat{\mathcal{H}}(\mathbf{r}, -i\nabla) \psi_m(\mathbf{r}, \mathbf{k}) \\ &\simeq \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) \hat{\mathcal{H}}_0(\mathbf{r}, -i\nabla) \psi_m(\mathbf{r}, \mathbf{k}) + e\mathcal{E}_0 \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) \hat{\Omega}(\mathbf{r}, \mathbf{k}) \psi_m(\mathbf{r}, \mathbf{k}) \end{aligned}$$

An energy function exists for crystals in a constant electric field $\mathcal{E}_0 = (\mathcal{E}_x, \mathcal{E}_y, \mathcal{E}_z)$ so that the energy of the perturbed system can be expressed in powers of the field components (denoted by indices t, u, v, w) as

$$E(\mathcal{E}) = E(0) - \sum_i \mu_t \mathcal{E}_t - \frac{1}{2!} \sum_{tu} \alpha_{tu} \mathcal{E}_t \mathcal{E}_u - \frac{1}{3!} \sum_{tuv} \beta_{tuv} \mathcal{E}_t \mathcal{E}_u \mathcal{E}_v - \frac{1}{4!} \sum_{tuvw} \gamma_{tuvw} \mathcal{E}_t \mathcal{E}_u \mathcal{E}_v \mathcal{E}_w + \dots$$

$$\mu_t = - \left. \frac{\partial E}{\partial \mathcal{E}_t} \right|_{\mathcal{E}_t=0}$$

$$\alpha_{tu} = - \left. \frac{\partial^2 E}{\partial \mathcal{E}_t \partial \mathcal{E}_u} \right|_{\mathcal{E}_t=\mathcal{E}_u=0}$$

$$\beta_{tuv} = - \left. \frac{\partial^3 E}{\partial \mathcal{E}_t \partial \mathcal{E}_u \partial \mathcal{E}_v} \right|_{\mathcal{E}_t=\mathcal{E}_u=\mathcal{E}_v=0}$$

$$\gamma_{tuvw} = - \left. \frac{\partial^4 E}{\partial \mathcal{E}_t \partial \mathcal{E}_u \partial \mathcal{E}_v \partial \mathcal{E}_w} \right|_{\mathcal{E}_t=\mathcal{E}_u=\mathcal{E}_v=\mathcal{E}_w=0}$$

represent optical properties of increasing order, respectively, the permanent electric dipole moment $\boldsymbol{\mu}$ (vector), the polarizability $\boldsymbol{\alpha}$ (tensor of rank 2), the first hyperpolarizability $\boldsymbol{\beta}$ (tensor of rank 3), and the second hyperpolarizability $\boldsymbol{\gamma}$ (tensor of rank 4).

Dielectric Tensor

Starting from the total energy E of the perturbed system expressed in powers of the electric field components, in the same way, the electric dipole moment components can be expanded in powers of the electric field components as

$$\mu_t(\mathcal{E}) = \mu_t(\mathbf{0}) + \sum_u \alpha_{tu} \mathcal{E}_u + \frac{1}{2!} \sum_{uv} \beta_{tuv} \mathcal{E}_u \mathcal{E}_v + \frac{1}{3!} \sum_{uvw} \gamma_{tuvw} \mathcal{E}_u \mathcal{E}_v \mathcal{E}_w + \dots$$

Using the previous expansion, together with the relation between the electric dipole moment and the induced polarization vector, the induced polarization vector components can be found to be

$$\begin{aligned} P_t(\mathcal{E}) &= \frac{1}{v} [\mu_t(\mathcal{E}) - \mu_t(\mathbf{0})] = \frac{1}{v} \sum_u \alpha_{tu} \mathcal{E}_u + \frac{1}{2!v} \sum_{uv} \beta_{tuv} \mathcal{E}_u \mathcal{E}_v + \frac{1}{3!v} \sum_{uvw} \gamma_{tuvw} \mathcal{E}_u \mathcal{E}_v \mathcal{E}_w + \dots \\ &= \epsilon_0 \sum_u \chi_{tu} \mathcal{E}_u + \frac{\epsilon_0}{2!} \sum_{uv} \chi_{tuv} \mathcal{E}_u \mathcal{E}_v + \frac{\epsilon_0}{3!} \sum_{uvw} \chi_{tuvw} \mathcal{E}_u \mathcal{E}_v \mathcal{E}_w + \dots \end{aligned}$$

where the first dielectric susceptibility components χ_{tu} have been defined as

$$\text{SI units : } \chi_{tu} = \frac{\alpha_{tu}}{v\epsilon_0} \quad \text{au units : } \chi_{tu} = \frac{4\pi}{v} \alpha_{tu} \quad \rightarrow \quad [\chi_{tu}] = \frac{[\alpha_{tu}]}{[L^3]} = \text{dimensionless}$$

Finally, in a three dimensional system, the dielectric tensor components are related to the corresponding components of the polarizability and the susceptibility as

$$\text{au units : } \epsilon_{tu} = \delta_{tu} + \frac{4\pi}{v} \alpha_{tu} = \delta_{tu} + \chi_{tu}$$

Linear Optical Response Properties in the CRYSTAL code

The input required to activate the calculation of linear optical properties through the CPHF or CPKS method, as implemented in the CRYSTAL code, is very simple and includes different optional keywords. Tutorial: <https://tutorials.crystalsolutions.eu/tutorial.html?td=cphf&tf=cphf>

[in the first block : nuclear geometry control block]

CPHF [CPKS]	Keyword to open the CPHF [CPKS] input block
FMIXING	percentage mixing of Fock matrix derivatives from iterations n and $n - 1$
50	[default: 0]
MAXCYCLE	maximum number of SC-CP cycles
100	[default: 100]
TOTALPHA	convergence threshold on $\Delta\alpha_{uu}$ between SC-CP iterations set to $10^{-\text{tol}}$
4	[default: tol = 3]
RESTART	restart CPHF [CPKS] from previous run
END	End of the CPHF [CPKS] input block

CPHF [CPKS]	The CRYSTAL code fully exploits symmetry for calculating the polarizability tensor components. In addition to this, the computational cost can be further reduced by selecting certain specific components to calculate. This can be done by replacing the optional keyword MAXCYCLE with the optional keyword SELEDIR .
FMIXING	
50	
SELEDIR	
100 0 0	
TOTALPHA	In the example given on the left panel, the maximum number of SC-CP iterations allowed along x is set to 100, while the number of iterations allowed along y and z is set to 0 (that is, α_{yy} and α_{zz} are calculated at the sum over states level).
4	
RESTART	
END	

Linear Optical Response Properties in the CRYSTAL code

The accuracy of the linear optical properties computed through the CPHF and CPKS approach with CRYSTAL relies on a number of different computational parameters.

One of these computational parameters is the convergence threshold on the self-consistent unperturbed total energy (keyword TOLDEE). It has been found (see Refs. [1,2]) that setting the TOLDEE value in the range (8, 10) is enough for assuring well-converged optical properties.

Analogously, the definition of the density functional theory integration grid is no more critical for the evaluation of the (hyper)polarizability than for total energy calculations. A change from the predefined smallest grid to the XLGRID implies an improvement smaller than 0.1%.

The computational parameters that deserve more attention in the calculation of the linear optical response properties are

- (i) the number of $\mathbf{k}$ points at which the Fock (Kohn-Sham) matrix is diagonalized in the first Brillouin zone (keyword **SHRINK**)
- (ii) the five thresholds to set the selection of the Coulomb and exact exchange integrals contributions to the total energy (keyword **TOLINTEG**)

Difficulties in stabilizing the linear optical properties with respect to both parameters strongly depend on the band gap width. This is a consequence of the relation of inverse proportionality between the energy gap and the energy derivatives that appears in the equation for the calculation of the polarizability. As a general rule, the smaller the estimated band gap, the slower the convergence.

[1] R. Orlando et al., J. Chem. Phys. 131, 184105 (2009)

[2] R. Orlando et al., J. Chem. Phys. 132, 244106 (2010)

Linear Optical Response Properties in the CRYSTAL code

Exercise 1.

Effect of the number of reciprocal space points $\mathbf{k}$ on the linear optical response properties

System : SiC crystal with $F\bar{4}3m$ symmetry and experimental lattice parameter 4.358 Å

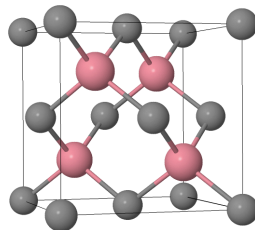
- Run different CPKS calculations with PBE and B3LYP functionals, varying the parameter SHRINK as given in the table, collecting the results for

↗ the polarizability tensor component α_{xx}

↗ the dielectric tensor component ϵ_{xx}

↗ the susceptibility tensor component χ_{xx}

- What is the minimum shrinking factor that provides well-converged optical properties (i.e. above which the calculated properties change by about 0.10% or less)?



		PBE functional			B3LYP functional		
SHRINK	$n_{\mathbf{k}}$	α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}	α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}
6	6						
8	8						
10	10						
12	12						
14	14						
16	16						

Linear Optical Response Properties in the CRYSTAL code

Exercise 1. Solution.

Effect of the number of reciprocal space points $\mathbf{k}$ on the linear optical response properties

System : SiC crystal with $F\bar{4}3m$ symmetry and experimental lattice parameter 4.358 Å

- A value of the SHRINK parameter equal to

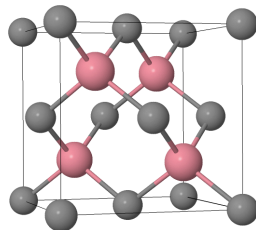
SHRINK 12 12

assures a convergence of all the optical properties

↔ by about 0.10% for the PBE functional

↔ by less than 0.10% for the B3LYP functional

with respect to the well converged properties associated to the higher value 16 16 of the SHRINK parameter



PBE functional						B3LYP functional		
SHRINK	$n_{\mathbf{k}}$		α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}	α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}
6 6	16		68.1214	7.1305	6.1305	57.4584	6.1709	5.1709
8 8	29		63.6536	6.7284	5.7284	55.0363	5.9529	4.9529
10 10	47		62.5777	6.6316	5.6316	54.5576	5.9098	4.9098
12 12	72		62.3216	6.6085	5.6085	54.4660	5.9016	4.9016
14 14	104		62.2610	6.6031	5.6031	54.4487	5.9000	4.9000
16 16	145		62.2467	6.6018	5.6018	54.4455	5.8997	4.8997

Linear Optical Response Properties in the CRYSTAL code

Exercise 2.

Effect of the bielectronic integrals thresholds on the linear optical response properties

System : SiC crystal with $F\bar{4}3m$ symmetry and experimental lattice parameter 4.358 Å

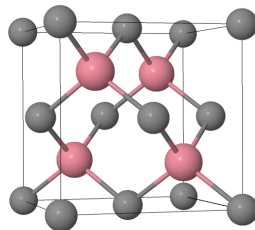
- Run different CPKS calculations with PBE and B3LYP functionals, varying the parameter TOLINTEG as given in the table, collecting the results for

↗ the polarizability tensor component α_{xx}

↗ the dielectric tensor component ϵ_{xx}

↗ the susceptibility tensor component χ_{xx}

- What is the minimum shrinking factor that provides well-converged optical properties (i.e. above which the calculated properties change by about 0.10% or less)?



TOLINTEG	PBE functional			B3LYP functional		
	α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}	α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}
7 7 7 7 14						
8 8 8 8 16						
10 10 10 10 20						
12 12 12 12 24						
14 14 14 14 28						
16 16 16 16 32						

Linear Optical Response Properties in the CRYSTAL code

Exercise 2. Solution.

Effect of the bielectronic integrals thresholds on the linear optical response properties

System : SiC crystal with $F\bar{4}3m$ symmetry and experimental lattice parameter 4.358 Å

- A value of the TOLINTEG parameter equal to

TOLINTEG 7 7 7 7 14 for PBE functional

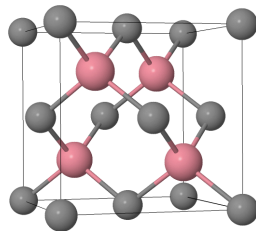
TOLINTEG 12 12 12 12 24 for B3LYP functional

assures a convergence of all the optical properties

↔ by about 0.01% for the PBE functional

↔ by less than 0.10% for the B3LYP functional

with respect to the well converged properties associated to the higher value of the TOLINTEG parameter



PBE functional				B3LYP functional		
TOLINTEG	α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}	α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}
7 7 7 7 14	62.3299	6.6093	5.6093	54.4945	5.9042	4.9042
8 8 8 8 16	62.3216	6.6085	5.6085	54.4660	5.9016	4.9016
10 10 10 10 20	62.3226	6.6086	5.6086	54.5542	5.9095	4.9095
12 12 12 12 24	62.3237	6.6087	5.6087	54.6018	5.9138	4.9138
14 14 14 14 28	62.3245	6.6088	5.6088	54.6646	5.9195	4.9195
16 16 16 16 32	62.3235	6.6087	5.6087	54.6798	5.9208	4.9208

Linear Optical Response Properties in the CRYSTAL code

Exercise 3.

Effect of the exchange-correlation functional on the linear optical response properties

System : SiC crystal with $F\bar{4}3m$ symmetry and experimental lattice parameter 4.358 Å

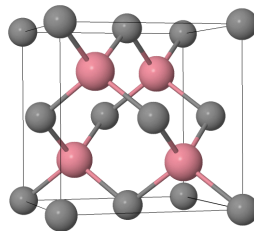
- Run different CPKS calculations with the previously found parameters SHRINK and TOLINTEG at convergence, varying the exchange-correlation functional as given in the table, collecting the results for

↗ the polarizability tensor component α_{xx}

↗ the dielectric tensor component ϵ_{xx}

↗ the susceptibility tensor component χ_{xx}

- What is the functional that closely reproduces the experimental band gap E_g and optical properties?



Exchange-correlation functional	E_g [eV]	α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}
LDA				
PBE				
PBE0				
B3LYP				
Experimental value	2.40 [1]		6.52 [2]	

[1] C. Persson et al., J. Appl. Phys. 86, 4419-4427 (1999)

[2] L. Patrick et al., Phys. Rev. B 2, 2255 (1970)

Linear Optical Response Properties in the CRYSTAL code

Exercise 3. Solution.

Effect of the exchange-correlation functional on the linear optical response properties

System : SiC crystal with $F\bar{4}3m$ symmetry and experimental lattice parameter 4.358 Å

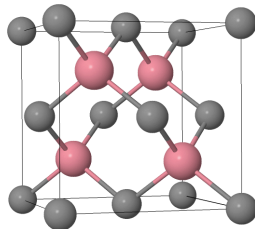
– The errors in reproducing the experimental data are

↷ $dE_g = 1.0889$ eV and $d\epsilon_{xx} = 1.5\%$ for LDA functional

↷ $dE_g = 1.0422$ eV and $d\epsilon_{xx} = 1.4\%$ for PBE functional

↷ $dE_g = 0.5126$ eV and $d\epsilon_{xx} = 8.1\%$ for PBE0 functional

↷ $dE_g = 0.5406$ eV and $d\epsilon_{xx} = 9.3\%$ for B3LYP functional



Exchange-correlation functional	E_g [eV]	α_{xx} [Bohr ³]	ϵ_{xx}	χ_{xx}
LDA	1.3111	62.4053	6.6161	5.6161
PBE	1.3578	62.3299	6.6093	5.6093
PBE0	2.9126	55.4749	5.9924	4.9924
B3LYP	2.9406	54.6018	5.9138	4.9138
Experimental value	2.40 [1]		6.52 [2]	

[1] C. Persson et al., J. Appl. Phys. 86, 4419-4427 (1999)

[2] L. Patrick et al., Phys. Rev. B 2, 2255 (1970)

Backup slides

Line integral of the electric field

In order to find the form of the scalar potential $\phi(\mathbf{r}, t)$ in the length gauge, it is required to solve the integral

$$\phi(\mathbf{r}, t) = - \int \boldsymbol{\mathcal{E}}(\mathbf{r}, t) \cdot d\mathbf{r} \quad \text{with} \quad \boldsymbol{\mathcal{E}}(\mathbf{r}, t) = \mathcal{E}_0 \hat{\mathbf{q}} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$$

that is a line integral of the vector field $\boldsymbol{\mathcal{E}}(\mathbf{r}, t)$ that results in a scalar function $\phi(\mathbf{r}, t)$. For the integration, it is possible to choose a straight line path in the direction of $\hat{\mathbf{q}}$ as this is the direction in which the electric field varies. The path can be parametrized as

$$\mathbf{r}(s) = s \hat{\mathbf{q}} \quad \rightarrow \quad d\mathbf{r}(s) = \hat{\mathbf{q}} ds$$

so that

$$\boldsymbol{\mathcal{E}}(s, t) \equiv \boldsymbol{\mathcal{E}}(\mathbf{r}(s), t) = \mathcal{E}_0 \hat{\mathbf{q}} e^{i(s\mathbf{q} \cdot \hat{\mathbf{q}} - \omega t)} = \mathcal{E}_0 \hat{\mathbf{q}} e^{i(sq\hat{\mathbf{q}} \cdot \hat{\mathbf{q}} - \omega t)} = \mathcal{E}_0 \hat{\mathbf{q}} e^{i(sq - \omega t)}$$

where the relations $\mathbf{q} = q \hat{\mathbf{q}}$ with $q = \|\mathbf{q}\|$ and $\hat{\mathbf{q}} \cdot \hat{\mathbf{q}} = 1$ have been used. Adopting this parametrization, the integral becomes

$$\begin{aligned} \phi(s, t) &= - \int \boldsymbol{\mathcal{E}}(s, t) \cdot \hat{\mathbf{q}} ds = -\mathcal{E}_0 \int (\hat{\mathbf{q}} \cdot \hat{\mathbf{q}}) e^{i(sq - \omega t)} ds \\ &= -\mathcal{E}_0 \int e^{i(sq - \omega t)} ds = -\frac{\mathcal{E}_0}{iq} e^{i(sq - \omega t)} + c(t) = \frac{i\mathcal{E}_0}{q} e^{i(sq - \omega t)} + c(t) \end{aligned}$$

where $c(t)$ is an integration constant (possibly time dependent)

Line integral of the electric field

$$\phi(s, t) = \frac{\imath \mathcal{E}_0}{q} e^{\imath(sq - \omega t)} + c(t)$$

where $c(t)$ is an integration constant (possibly time dependent). Since s is a scalar variable along the direction of $\hat{\mathbf{q}}$ and it can be written using the parametrization relation as

$$\mathbf{r}(s) = s \hat{\mathbf{q}} \quad \rightarrow \quad \hat{\mathbf{q}} \cdot \mathbf{r} = s \hat{\mathbf{q}} \cdot \hat{\mathbf{q}} = s \quad \rightarrow \quad s = \hat{\mathbf{q}} \cdot \mathbf{r}$$

the final form of the scalar potential $\phi(\mathbf{r}, t)$ can be derived substituting this last relation in the expression of $\phi(s, t)$ that results from the previously derived solution of the line integral, thus leading to the expression

$$\phi(\mathbf{r}, t) = \frac{\imath \mathcal{E}_0}{q} e^{\imath(q\hat{\mathbf{q}} \cdot \mathbf{r} - \omega t)} + c(t) = \frac{\imath \mathcal{E}_0}{q} e^{\imath(\mathbf{q} \cdot \mathbf{r} - \omega t)} + c(t)$$

The integration constant $c(t)$ can possibly be time dependent. However, choosing a time independent value equal to $-\imath \mathcal{E}_0/q$ the form of the scalar potential becomes

$$c(t) = -\frac{\imath \mathcal{E}_0}{q} \quad \rightarrow \quad \phi(\mathbf{r}, t) = \frac{\imath \mathcal{E}_0}{q} \left[e^{\imath(\mathbf{q} \cdot \mathbf{r} - \omega t)} - 1 \right]$$

This choice of $c(t)$ makes the scalar potential vanish at the origin, that is for $\mathbf{r} = \mathbf{0}$ and $t = 0$ the scalar potential $\phi(\mathbf{r} = \mathbf{0}, t = 0) = 0$ is equal to zero.

Scalar wave equation for a pure radiative field

Consider the scalar wave equation for a pure radiative field that can be derived from the Maxwell equations applied in the vacuum (with the charged source being placed far away from the electron), that is

$$\nabla^2 \phi(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2 \phi(\mathbf{r}, t)}{\partial t^2} = 0 \quad (1)$$

where $\phi(\mathbf{r}, t)$ is a scalar field (generally called scalar potential). Given the wave vector $\mathbf{q}$ and the angular frequency ω , a solution to the scalar wave equation (1) for a pure radiative field is

$$\phi(\mathbf{r}, t) = \frac{1}{q} \left[e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} - 1 \right] \quad (2)$$

Proof. To prove that the form (2) of the scalar field is a solution of the wave equation (1), the procedure is to substitute the scalar field form (2) in the differential equation (1) to demonstrate that the wave equation form (1) is satisfied. Consider first the spatial derivatives

$$\nabla^2 \phi(\mathbf{r}, t) = \nabla \cdot \nabla \phi(\mathbf{r}, t) = \frac{1}{q} \nabla \cdot \left[i\mathbf{q} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} \right] = -\frac{1}{q} (\mathbf{q} \cdot \mathbf{q}) e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} = -i q e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$$

where, in the last identity, the relation $\mathbf{q} \cdot \mathbf{q} = \|\mathbf{q}\|^2 = q^2$ has been used. At the same time, the part related to the time derivatives leads to the term

$$\frac{1}{c^2} \frac{\partial^2 \phi(\mathbf{r}, t)}{\partial t^2} = \frac{1}{qc^2} \frac{\partial}{\partial t} \left[-i\omega e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} \right] = -i \frac{\omega^2}{qc^2} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$$

Scalar wave equation for a pure radiative field

Considering the previously derived spatial and time derivatives, that are

$$\nabla^2 \phi(\mathbf{r}, t) = -i q e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} \quad \text{and} \quad \frac{1}{c^2} \frac{\partial^2 \phi(\mathbf{r}, t)}{\partial t^2} = -i \frac{\omega^2}{q c^2} e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$$

and substituting these two derivatives forms in the wave equation (1), it becomes

$$\nabla^2 \phi(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2 \phi(\mathbf{r}, t)}{\partial t^2} = -i \left(q - \frac{\omega^2}{q c^2} \right) e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} = 0 \quad (3)$$

Equation (3) is therefore satisfied if and only if the following relation holds

$$q - \frac{\omega^2}{q c^2} = 0 \quad \rightarrow \quad \omega^2 = q^2 c^2 \quad (4)$$

Therefore, it has been proved that if and only if the relation

$$\omega = q c \quad (5)$$

is satisfied, the form (2) for the scalar field $\phi(\mathbf{r}, t)$ is a solution of the wave equation (1) for a pure radiative field. $\square$

Hamiltonian matrix elements

$$\hat{\mathcal{H}}_{\phi}^{nm}(\mathbf{k}) = \lim_{\mathbf{q} \rightarrow \mathbf{0}} \hat{\mathcal{H}}_{\phi}^{nm}(\mathbf{k}, \mathbf{k} - \mathbf{q}) = - \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left\{ \frac{i e \mathcal{E}_0}{q} \left[\int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) e^{i\mathbf{q} \cdot \mathbf{r}} \psi_m(\mathbf{r}, \mathbf{k} - \mathbf{q}) - \delta(\mathbf{q}) \delta_{nm} \right] \right\}$$

In the case $\mathbf{q} \rightarrow \mathbf{0}$ if compared to $\mathbf{k}$, the wavefunction $\psi_n(\mathbf{r}, \mathbf{k} - \mathbf{q})$ can be expanded as a Taylor series truncated to the first order around $\mathbf{k}$ as

$$\psi_n(\mathbf{r}, \mathbf{k} - \mathbf{q}) \simeq \psi_n(\mathbf{r}, \mathbf{k}) - \mathbf{q} \cdot \nabla_{\mathbf{k}} \psi_n(\mathbf{r}, \mathbf{k})$$

and similarly for the exponential operator

$$e^{i\mathbf{q} \cdot \mathbf{r}} \simeq 1 + i\mathbf{q} \cdot \mathbf{r}$$

In the following, the explicit expression for the perturbative Hamiltonian matrix elements is found, carrying out the integrations and doing the limits.

Inserting the previous two approximations in the first term of the Hamiltonian matrix element, it becomes

$$\begin{aligned} \hat{\mathcal{F}}_{nm}(\mathbf{k}) &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left[\frac{i e \mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) e^{i\mathbf{q} \cdot \mathbf{r}} \psi_m(\mathbf{r}, \mathbf{k} - \mathbf{q}) \right] \\ &\simeq \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left\{ \frac{i e \mathcal{E}_0}{q} \left[\int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) \psi_m(\mathbf{r}, \mathbf{k}) - \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{q} \cdot \nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right. \right. \\ &\quad \left. \left. + i \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{q} \cdot \mathbf{r}) \psi_m(\mathbf{r}, \mathbf{k}) - i \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{q} \cdot \mathbf{r}) (\mathbf{q} \cdot \nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right] \right\} \end{aligned}$$

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$$\begin{aligned}
 \hat{\mathcal{F}}_{nm}(\mathbf{k}) &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left[\frac{\imath e \mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) e^{\imath \mathbf{q} \cdot \mathbf{r}} \psi_m(\mathbf{r}, \mathbf{k} - \mathbf{q}) \right] \\
 &\simeq \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left\{ \frac{\imath e \mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) \psi_m(\mathbf{r}, \mathbf{k}) - \frac{\imath e \mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{q} \cdot \nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right. \\
 &\quad \left. - \frac{e \mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{q} \cdot \mathbf{r}) \psi_m(\mathbf{r}, \mathbf{k}) + \frac{e \mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{q} \cdot \mathbf{r}) (\mathbf{q} \cdot \nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right\}
 \end{aligned}$$

The first term in $\hat{\mathcal{F}}_{nm}(\mathbf{k})$ can be computed as

$$\hat{\mathcal{P}}_{nm}(\mathbf{k}) = \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left[\frac{\imath e \mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) \psi_m(\mathbf{r}, \mathbf{k}) \right] = \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left(\frac{\imath e \mathcal{E}_0}{q} \delta_{nm} \right)$$

The second and the third terms in $\hat{\mathcal{F}}_{nm}(\mathbf{k})$ can be collected in the form

$$\begin{aligned}
 \hat{\mathcal{Q}}_{nm}(\mathbf{k}) &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left[\frac{e \mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{q} \cdot \mathbf{r}) \psi_m(\mathbf{r}, \mathbf{k}) + \frac{\imath e \mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{q} \cdot \nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right] \\
 &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left\{ e \mathcal{E}_0 \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) [(\mathbf{u}_q \cdot \mathbf{r}) + \imath (\mathbf{u}_q \cdot \nabla_{\mathbf{k}})] \psi_m(\mathbf{r}, \mathbf{k}) \right\} \quad \text{with} \quad \mathbf{u}_q = \mathbf{q} / \|\mathbf{q}\| = \mathbf{q} / q \\
 &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left\{ e \mathcal{E}_0 \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) [\mathbf{u}_q \cdot (\mathbf{r} + \imath \nabla_{\mathbf{k}})] \psi_m(\mathbf{r}, \mathbf{k}) \right\} \\
 &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left\{ e \mathcal{E}_0 \mathbf{u}_q \cdot \left[\int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{r} + \imath \nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right] \right\} \\
 &= e \mathcal{E}_0 \mathbf{u}_q \cdot \left[\int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{r} + \imath \nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right] \quad \text{where} \quad \mathcal{E}_0 = \mathcal{E}_0 \mathbf{u}_q
 \end{aligned}$$

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The third term in $\hat{\mathcal{F}}_{nm}(\mathbf{k})$ can be computed as

$$\begin{aligned}\hat{\mathcal{R}}_{nm}(\mathbf{k}) &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left[\frac{e\mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{q} \cdot \mathbf{r}) (\mathbf{q} \cdot \nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right] \quad \text{where} \quad \mathbf{q} = \|\mathbf{q}\| \mathbf{u}_q = q \mathbf{u}_q \\ &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left[e\mathcal{E}_0 q \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{u}_q \cdot \mathbf{r}) (\mathbf{u}_q \cdot \nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right] = 0\end{aligned}$$

Therefore, collecting the three terms computed, the operator $\hat{\mathcal{F}}_{nm}(\mathbf{k})$ can be written as

$$\begin{aligned}\hat{\mathcal{F}}_{nm}(\mathbf{k}) &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left[\frac{ie\mathcal{E}_0}{q} \int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) e^{i\mathbf{q} \cdot \mathbf{r}} \psi_m(\mathbf{r}, \mathbf{k} - \mathbf{q}) \right] \\ &\simeq \hat{\mathcal{P}}_{nm}(\mathbf{k}) + \hat{\mathcal{Q}}_{nm}(\mathbf{k}) + \hat{\mathcal{R}}_{nm}(\mathbf{k}) \\ &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left(\frac{ie\mathcal{E}_0}{q} \delta_{nm} \right) - e\mathcal{E}_0 \cdot \left[\int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{r} + i\nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right]\end{aligned}$$

where $\mathcal{E}_0 = \mathcal{E}_0 \mathbf{u}_q = \mathcal{E}_0 \mathbf{q} / \|\mathbf{q}\| = \mathcal{E}_0 \mathbf{q} / q$ with $\mathbf{u}_q$ a unit vector. Finally, inserting this expression for $\hat{\mathcal{F}}_{nm}(\mathbf{k})$ back to the original perturbative Hamiltonian element, it becomes

$$\begin{aligned}\hat{\mathcal{H}}_{\phi}^{nm}(\mathbf{k}) &= \lim_{\mathbf{q} \rightarrow \mathbf{0}} \hat{\mathcal{H}}_{\phi}^{nm}(\mathbf{k}, \mathbf{k} - \mathbf{q}) = -\hat{\mathcal{F}}_{nm}(\mathbf{k}) + \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left[\frac{ie\mathcal{E}_0}{q} \delta(\mathbf{q}) \delta_{nm} \right] \\ &\simeq e\mathcal{E}_0 \cdot \left[\int_{-\infty}^{\infty} d\mathbf{r} \, \psi_n^*(\mathbf{r}, \mathbf{k}) (\mathbf{r} + i\nabla_{\mathbf{k}}) \psi_m(\mathbf{r}, \mathbf{k}) \right] - \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left(\frac{ie\mathcal{E}_0}{q} \delta_{nm} \right) + \lim_{\mathbf{q} \rightarrow \mathbf{0}} \left[\frac{ie\mathcal{E}_0}{q} \delta(\mathbf{q}) \delta_{nm} \right]\end{aligned}$$

In the perturbative approach, only the terms with $n \neq m$ are considered, so that the perturbative Hamiltonian matrix element reduces to the first term only in the right hand side of the last expression.